Макаровский Кирилл.

№18.

$\sigma(z) = \frac{1}{1+e^{-z}}$, $\sigma' \overset{?}{=} \sigma(1-\sigma)$

$$\sigma' = \frac{e^{-z}}{(1+e^{-z})^2} = \frac{1}{(1+e^{-z})} \cdot \frac{e^{-z}}{(1+e^{-z})} = \frac{1}{1+e^{-z}}\left(\frac{1-1+e^{-z}}{1+e^{-z}}\right) =$$

$$= \frac{1}{1+e^{-z}}\left(1 - \frac{1}{1+e^{-z}}\right) = \sigma(1-\sigma) \quad \boxtimes$$

№19

$\{1, 2, \dots, K\}$ - класы

$$g_k(s_1, \dots, s_K) = \frac{e^{s_k}}{\sum_{i=1}^{K} e^{s_i}}$$ – softmax function.

$$R^{(j)} = -\sum_{i=1}^{K} I(y^{(j)} = i) \ln g_i(s_1, \dots, s_K)$$

Д-ть:

1) $\frac{\partial g_k}{\partial s_l} = g_k(I(k=l) - g_l)$

$$\frac{\partial g_k}{\partial s_l} = \frac{e^{s_k}}{\sum_{l=1}^{K} e^{s_l}} - \frac{e^{s_k} e^{s_l}}{\left(\sum_{l=1}^{K} e^{s_l}\right)^2} = \frac{e^{s_k}}{\sum_{l=1}^{K} e^{s_l}}\left(1 - \frac{e^{s_l}}{\sum_{l=1}^{K} e^{s_l}}\right) =$$

$= g_k(1 - g_l)$, $k = l$

$$\frac{\partial g_k}{\partial s_l} = \frac{-e^{s_k} e^{s_l}}{\left(\sum_{l=1}^{K} e^{s_l}\right)^2} = -g_k \cdot g_l, \quad k \neq l;$$

$$I(k=l) = \begin{cases} 1 & k=l \\ 0 & k \neq l \end{cases}$$

2) $$\frac{\partial R^{(i)}}{\partial g_k} = -\frac{I(y^{(i)})}{g_k}.$$

$$\frac{\partial R^{(i)}}{\partial g_k} = -I(y^{(i)}=k) \cdot \frac{1}{g_k(s_1, \dots, s_k)} = -\frac{I(y^{(i)})}{g_k}.$$

3) $$\frac{\partial R^{(i)}}{\partial s_l} = g_l - I(l=y^{(i)})$$

$$\frac{\partial R^{(i)}}{\partial s_l} = \sum_{l=1}^{K} \frac{\partial R^{(i)}}{\partial g_k} \cdot \frac{\partial g_k}{\partial s_l} = \sum_{l=1}^{K} \left(-\frac{I(y^{(i)}=k)}{g_k(s_1, \dots, s_k)} \right) \cdot$$

$$\cdot\ g_k \left(I(k=l) - g_l \right) = -\left(I(y^{(i)}=l) - g_l \right) =$$

$$= g_l - I(l=y^{(i)})$$